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Unidades: E [MPa] · I [m⁴] · L [m] · M [kN·m] · V [kN] · θ [rad] · w [kN/m] · P [kN]

1. DATOS DE ENTRADA

Nodos

ID	Etiqueta	Apoyo ($\theta=0$)	G.L.
1	A	Sí	θ_A
2	B	No	θ_B
3	C	No	θ_C
4	D	Sí	θ_D

Elementos

Elem.	Ni	Nj	E (MPa)	I (m ⁴)	L (m)
1	A	B	210000	0.00015	5
2	B	C	210000	0.00015	4
3	C	D	210000	0.00015	5

Cargas en elementos

ID	Elem.	Tipo	Valor	Posición (m)
1	E1	w (kN/m)	12	—
2	E2	P (kN)	25	2
3	E3	w (kN/m)	8	—

Momentos nodales (kN·m)

Nodo	M
—	θ

2. MATRICES DE RIGIDEZ LOCALES

$E1 \cdot L = 5 \text{ m} \cdot EI/L = 6.3000$

$$k1 = \begin{bmatrix} & \theta_A & \theta_B \\ \theta_A & 25.2000 & 12.6000 \\ \theta_B & 12.6000 & 25.2000 \end{bmatrix}$$

$E2 \cdot L = 4 \text{ m} \cdot EI/L = 7.8750$

$$k2 = \begin{bmatrix} & \theta_B & \theta_C \\ \theta_B & 31.5000 & 15.7500 \\ \theta_C & 15.7500 & 31.5000 \end{bmatrix}$$

$$E3 \cdot L = 5 \text{ m} \cdot EI/L = 6.3000$$

$$k3 = \begin{bmatrix} & \theta C & \theta D \\ \theta C & 25.2000 & 12.6000 \\ \theta D & 12.6000 & 25.2000 \end{bmatrix}$$

3. FUERZAS DE EMPOTRAMIENTO

[kN·m]

$$Fk1 = \begin{bmatrix} \theta A & 25.0000 \\ \theta B & -25.0000 \end{bmatrix}$$

[kN·m]

$$Fk2 = \begin{bmatrix} \theta B & 12.5000 \\ \theta C & -12.5000 \end{bmatrix}$$

[kN·m]

$$Fk3 = \begin{bmatrix} \theta C & 16.6667 \\ \theta D & -16.6667 \end{bmatrix}$$

4. VECTOR DE CARGA

$$Fk = F_{nodal} - F_{ki} \quad [kN \cdot m]$$

$$Fk = \begin{bmatrix} \theta A & -25.0000 \\ \theta B & 12.5000 \\ \theta C & -4.1667 \\ \theta D & 16.6667 \end{bmatrix}$$

G.L. libres · [kN·m]

$$Fd = \begin{bmatrix} \theta B & 12.5000 \\ \theta C & -4.1667 \end{bmatrix}$$

5. ENSAMBLE DE LA MATRIZ DE RIGIDEZ

Matriz global ensamblada (N×N)

$$KTG = \begin{bmatrix} & \theta A & \theta B & \theta C & \theta D \\ \theta A & 25.2000 & 12.6000 & 0 & 0 \\ \theta B & 12.6000 & 56.7000 & 15.7500 & 0 \\ \theta C & 0 & 15.7500 & 56.7000 & 12.6000 \\ \theta D & 0 & 0 & 12.6000 & 25.2000 \end{bmatrix}$$

$K_{11} = K_{dd}$ (G.L. libres)

$$K_{11} = \begin{bmatrix} & \theta_B & \theta_C \\ \theta_B & 56.7000 & 15.7500 \\ \theta_C & 15.7500 & 56.7000 \end{bmatrix}$$

Inversa de K_{11}

$$K_{11}^{-1} = \begin{bmatrix} & \theta_B & \theta_C \\ \theta_B & 0.0191 & -0.0053 \\ \theta_C & -0.0053 & 0.0191 \end{bmatrix}$$

$$K_{rd} = \begin{bmatrix} & \theta_B & \theta_C \\ \theta_A & 12.6000 & 0 \\ \theta_D & 0 & 12.6000 \end{bmatrix}$$

6. DESPLAZAMIENTOS

$D = K_{11}^{-1} \cdot F_d \cdot [\text{rad}]$

$$D = \begin{bmatrix} \theta_B & 0.2610 \\ \theta_C & -0.1460 \end{bmatrix}$$

7. FUERZAS INTERNAS Y REACCIONES

$F_i = k_i \cdot D_i + F_{ki} \cdot V = 1.9732 \text{ kN} \cdot [\text{kN}\cdot\text{m}]$

$$F_1 = \begin{bmatrix} M_{\theta A} & 28.2887 \\ M_{\theta B} & -18.4225 \end{bmatrix}$$

$F_i = k_i \cdot D_i + F_{ki} \cdot V = 1.3587 \text{ kN} \cdot [\text{kN}\cdot\text{m}]$

$$F_2 = \begin{bmatrix} M_{\theta B} & 18.4225 \\ M_{\theta C} & -12.9877 \end{bmatrix}$$

$F_i = k_i \cdot D_i + F_{ki} \cdot V = -1.1037 \text{ kN} \cdot [\text{kN}\cdot\text{m}]$

$$F_3 = \begin{bmatrix} M_{\theta C} & 12.9877 \\ M_{\theta D} & -18.5061 \end{bmatrix}$$

Elem.	Mi (kN·m)	Mj (kN·m)	V (kN)
E1	28.2887	-18.4225	1.9732
E2	18.4225	-12.9877	1.3587
E3	12.9877	-18.5061	-1.1037

$$FR = K_{rd} \cdot D + F_{rd} \cdot \theta \quad [\text{kN}\cdot\text{m}]$$

$$FR = \begin{bmatrix} \theta_A & -21.7113 \\ \theta_D & 14.8272 \end{bmatrix}$$